Jeffrey Bain Senior Software Engineer

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SUMMARY

Senior Software Engineer with 9+ years of experience building AI-powered, distributed, and real-time platforms at enterprise scale across fintech, healthcare, e-commerce, and logistics. Proven track record delivering high-performance systems at Stripe, Teladoc Health, eBay, and Uber, spanning LLM integrations, fraud prevention, healthcare interoperability, real-time communications, distributed microservices, and large-scale frontend systems. Strong expertise in Python, Go, TypeScript, React, Java, distributed architectures, and cloud-native system design. Experienced in building production AI systems, low-latency applications, and highly scalable customer-facing platforms.

PROFESSIONAL EXPERIENCE

Stripe, *Senior Software Engineer* 01/2023 – 05/2026 | Remote (United States)

Project: **Stripe Radar Assistant** (AI-Powered Fraud Rule Generation Platform)

- Built core AI infrastructure powering **Stripe Radar Assistant**, an LLM-driven fraud prevention copilot enabling merchants to generate complex fraud rules through natural language prompts.
- Developed Python services integrating **OpenAI GPT models** for translating merchant intent into executable Stripe Radar syntax.
- Designed and implemented **LLM validation guardrails** using Python, regex, and syntax validation pipelines to prevent malformed fraud rules from reaching production.
- Built real-time streaming experiences using **React, TypeScript, and Node.js**, enabling merchants to receive AI-generated rule recommendations interactively.
- Developed **backtesting systems** in Go and Ruby to evaluate AI-generated fraud rules against historical transaction datasets within seconds.
- Implemented **prompt injection protection mechanisms** and secure validation pipelines for production-grade AI safety.
- Built **context-aware recommendation systems** that personalized fraud rule suggestions using merchant-specific fraud history.
- Optimized LLM workflows and prompt pipelines, reducing inference/token costs by **40%** while improving generation quality.
- Contributed to the evolution of Radar Assistant into a **multi-turn conversational AI agent** for iterative fraud rule refinement.

Teladoc Health, *Senior Full Stack Engineer* 03/2020 – 12/2022 | Remote (United States)

Project: **Provider Consultation & Clinical Workflow Platform**

- Developed secure healthcare platform systems supporting **real-time provider-patient virtual consultations** under strict HIPAA compliance requirements.
- Built **FHIR/HL7-based API integrations** with healthcare systems including Epic and Cerner to securely retrieve and synchronize patient medical histories.
- Designed and improved backend services using **Java (Spring Boot), Go, PostgreSQL, and Redis** to support physician scheduling and clinical workflows.
- Re-engineered provider-patient matching systems, significantly improving consultation assignment speed and operational efficiency.
- Built **WebRTC and WebSocket failover mechanisms** to maintain stable video consultations for hospitals and clinical providers.
- Developed clinician-facing experiences using **React and TypeScript**, including split-screen workflows for simultaneous chart review and live consultation.
- Integrated **ambient AI documentation systems** powered by Azure OpenAI to generate real-time clinical summaries during patient visits.
- Implemented secure PHI masking, encryption, and compliance workflows to satisfy strict medical privacy standards.
- Mentored junior engineers and participated in architecture reviews and cross-functional technical planning.

eBay, *Full Stack Engineer* 07/2017 – 02/2020 | San Jose, CA

Project**: eBay Image Clean-Up (AI Seller Imaging Platform)**

- Built interactive browser-based image editing tools for **eBay's AI-powered image cleanup platform**, enabling sellers to optimize marketplace product imagery.

- Developed **HTML5 Canvas-based editing tools** allowing manual foreground/background correction for automated image segmentation.
- Improved React application performance for large image rendering, significantly reducing lag when processing high-resolution images.
- Resolved complex **cross-browser rendering issues** across Chrome, Safari, and Firefox.
- Modularized Android image-processing components into isolated libraries to improve maintainability and build performance.
- Implemented **JNI bridges between Kotlin/Java and high-performance C++ image processing engines**.
- Fixed critical mobile memory issues caused by high-resolution camera image handling.
- Optimized mobile AI inference workflows leveraging **device neural accelerators** to reduce image processing time to under one second.
- Implemented quality heuristics to determine when automatic cleanup confidence was insufficient and manual correction should be suggested.

Uber, *Software Engineer* 06/2016 – 06/2017 | San Francisco, CA

Project**: Uber Freight Marketplace Platform**

- Built distributed backend systems supporting **real-time freight marketplace coordination and shipment tracking**.
- Developed Go microservices for automated trucking carrier onboarding and insurance verification workflows.
- Designed PostgreSQL schemas for truck specifications and logistics matching systems.
- Optimized **Kafka-based event pipelines** for high-frequency truck GPS updates and shipment status tracking.
- Integrated **H3 geospatial indexing** to group freight activity into optimized logistics and pricing regions.
- Developed automated route monitoring and delay detection systems for operational visibility.
- Built shipment tracking dashboards using **React and Mapbox**, enabling enterprise users to monitor deliveries in real time.
- Developed Node.js APIs for streaming ETA updates and shipment state synchronization.

EDUCATION

Computer Science, *The University of Georgia* 08/2012 – 05/2016 | Athens, GA

- Focused on software engineering, distributed systems, algorithms, and system design.
- Built strong foundations in scalable architectures, backend systems, and performance optimization.

PROJECTS

AI-Powered Fraud Rule Generation Platform (Stripe Radar Assistant)

- Designed an LLM-powered platform converting natural language prompts into executable fraud detection rules.
- Built validation systems ensuring syntactic correctness and production safety of generated rules.
- Enabled merchants to test fraud rules against historical transaction datasets in seconds.

Real-Time Healthcare Consultation Platform

Built secure healthcare workflows supporting real-time virtual consultations, patient record retrieval, and AI-powered clinical note generation.

SKILLS

Languages (Python ▪ Go ▪ TypeScript ▪ JavaScript ▪ Java ▪ Ruby ▪ SQL ▪ Kotlin ▪ Swift ▪ C++)

Frontend (React ▪ TypeScript ▪ HTML5 Canvas ▪ WebSockets ▪ Node.js ▪ Mapbox)

Backend & APIs (Spring Boot ▪ Microservices ▪ REST APIs ▪ GraphQL ▪ Distributed Systems ▪ Event-Driven Architecture)

AI / Machine Learning (OpenAI API ▪ LLM Integration ▪ Prompt Engineering ▪ AI Safety ▪ Azure OpenAI ▪ NLP Systems)

Real-Time Systems (WebRTC ▪ Kafka ▪ Streaming Systems ▪ Low-Latency Systems ▪ Real-Time Data Processing)

Cloud & Infrastructure (AWS ▪ Docker ▪ Redis ▪ PostgreSQL ▪ Observability ▪ Monitoring ▪ Fault Tolerance)

Healthcare / Enterprise (FHIR ▪ HL7 ▪ HIPAA Compliance ▪ PHI Encryption ▪ EHR Integrations)